

		$v_A = 2.97 = 3.0 \text{ m s}^{-1}$
4	C	Let velocity before collision be v, and after collision be V. Then conservation of momentum gives $mv + 0 = 2mV$, giving $V = \frac{1}{2}v$. Initial kinetic energy $E_k = \frac{1}{2}mv^2$ and after collision, kinetic energy of the system is $\frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = \frac{1}{4}mv^2$. so change in kinetic energy is $\frac{1}{4}mv^2 - \frac{1}{2}mv^2 = -\frac{1}{4}mv^2 = -\frac{E_k}{2}$
5	C	Acceleration along the slope = $\frac{30-0}{4}$